Phishing Detection & Awareness Report

1. Introduction

Phishing is a social engineering attack in which attackers impersonate legitimate organizations to deceive users into revealing sensitive information such as usernames, passwords, OTPs, or financial details. These attacks are typically delivered through email and rely on psychological manipulation rather than technical exploitation.Phishing remains one of the most common causes of credential compromise and account takeover incidents in organizations. The objective of this report is to analyze a sample phishing email, identify key phishing indicators, classify the risk level, and provide preventive awareness guidelines for employees.

2. Email Sample Overview

Subject: ⚠ Urgent: Your Account Will Be Locked
Sender: security-support@secure-login-alert.com

Email Body Summary:

The email claims that suspicious activity has been detected on the user's account and requests immediate verification through a provided link. It warns that failure to act within 24 hours will result in permanent account suspension.

The link provided in the email is: `http://secure-account-verify[.]com`

3. Technical Analysis

3.1 Sender Domain Analysis

The sender email address is: `security-support@secure-login-alert.com`

Observations:

- The domain does not belong to any known legitimate organization.
- The domain name contains generic security-related keywords such as “secure” and “login”.
- There is no official company branding in the email.
- The sender identity appears spoofed or fabricated.

This indicates a high probability of phishing.

3.2 URL Analysis

The URL provided in the email is: `http://secure-account-verify[.]com`

Observations:

- The URL uses HTTP instead of HTTPS.
- The domain name is generic and not associated with any known organization.
- The domain contains keywords like “account” and “verify,” commonly used in phishing campaigns.
- The domain structure does not match any official service provider.

The browser blocked access to the domain, which further indicates potential malicious or invalid configuration.

Conclusion: The URL is highly suspicious.

3.3 Language & Psychological Indicators

The email uses several psychological manipulation techniques:

- Urgency: “Verify immediately”
- Fear: “Permanent account lock”
- Time pressure: “Within 24 hours”
- Generic greeting: “Dear User”

These tactics are commonly used in phishing attacks to trigger panic and reduce critical thinking.

4. Identified Phishing Indicators

The following phishing indicators were identified:

- Suspicious sender domain
- Generic greeting (no personalized name)
- Urgency-based messaging
- Fear-inducing language
- Suspicious and unverified URL
- Use of HTTP instead of HTTPS
- No official branding or contact information

Multiple indicators confirm malicious intent.

5. Risk Classification

Risk Level: **High**
Threat Category: **Phishing Attack**
Attack Objective: **Credential Harvesting**

Reason:The email attempts to redirect the user to a fake verification page where login credentials would likely be captured by the attacker.

6. How the Attack Works

1. The attacker sends a phishing email impersonating a legitimate organization.
2. The victim clicks on the malicious link.
3. A fake login page appears that looks legitimate.
4. The victim enters credentials.
5. The attacker collects the login information.
6. The attacker may perform account takeover or financial fraud.

This method is commonly used in banking and corporate credential theft campaigns.

7. Prevention Guidelines

For Employees

- Always verify the sender's email domain.
- Hover over links before clicking.
- Do not respond to urgent or threatening messages immediately.
- Verify suspicious emails with the IT or security team.
- Never enter credentials through email links.

Do's and Don'ts

Do:

- Report suspicious emails immediately.
- Check spelling and grammar errors.
- Confirm with official websites directly.

Don't:

- Click unknown links.
- Download unexpected attachments.
- Share passwords or OTPs via email.

8. Conclusion

The analyzed email exhibits multiple strong phishing indicators including a suspicious sender domain, malicious URL structure, urgency-based messaging, and fear tactics. Based on the technical and contextual analysis, the email is classified as a high-risk phishing attempt aimed at credential harvesting.

Employee awareness and proactive verification practices are essential to preventing such attacks.